HOMEWORK 6.10 · UNIT 6 · NC.8.F.3 · NC.8.F.4 · NC.8.F.2

Standard Form & Point-Slope Form

Targets: F.3-3, F.4-4, F.2-3, F.4-6, F.3-4, F.4-8, F.3-9

Name _____ Date _____ Block _____

Remember: Standard form: $Ax + By = C$. **Point-slope form:** $y - y_1 = m(x - x_1)$, where m is the slope and (x_1, y_1) is a point. To change to $y = mx + b$, **solve for y**. Ex: $2x + y = 5 \rightarrow y = -2x + 5$. $y - 3 = 2(x - 1) \rightarrow y - 3 = 2x - 2 \rightarrow y = 2x + 1$.

SCORE 3.0 Approaching · the basics

Targets: F.3-3, F.4-4, F.2-3

1 Which form is $5x - 2y = 10$? F.3-3

Slope-intercept | Standard | Point-slope

2 Write $2x + y = 9$ in slope-intercept form. F.4-4

3 Write $y - 1 = 4(x - 2)$ in slope-intercept form. F.4-4

Hint: Distribute first.

4 Which equation fits the table? REVIEW 6.8 F.4-4

x	0	1	2	3
y	1	6	11	16

(A) $y = 5x + 1$
(B) $y = x + 5$
(C) $y = 5x$
(D) $y = -5x + 1$

5 $y = 2x + 9$ grows faster than $y = 5x$. REVIEW 6.9 F.2-3

True | False

SCORE 4.0 Proficient · on target

Targets: F.4-6, F.3-4

6 A line has slope −3 and passes through (1, 5). Write its equation in point-slope form. Then write it in slope-intercept form. F.4-6

7 **Ana** made a mistake. Circle it. Explain it. Fix it. F.4-6

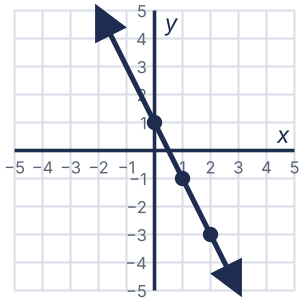
ANA'S WORK

$6x + 2y = 8$

$2y = -6x + 8$

$y = -6x + 4$

8 Which equation matches the graph? REVIEW 6.7 F.3-4



Ⓐ $y = -2x + 1$ Ⓑ $y = 2x + 1$ Ⓒ $y = x - 2$ Ⓓ $y = -\frac{1}{2}x + 1$

SCORE 5.0 Mastery · go deeper

Targets: F.4-8, F.3-9

9 A line passes through (−1, 2) and (3, 10). Write its equation in point-slope form and in slope-intercept form. Explain. F.4-8

m = _____ . Point-slope: _____ . Slope-intercept: _____ .

10 Mia says $y - 3 = 2(x - 1)$ and $y - 7 = 2(x - 3)$ are the same line. Is she right? Explain. F.3-9

Mia is _____ . In slope-intercept form, both are _____ .